Project Abstracts

Two Proposed Final-Year Projects

Abstract 1

An Intelligent Algorithm-Selection Framework for Resource-Constrained Project Scheduling Using Neural Networks

The Resource-Constrained Project Scheduling Problem (RCPSP) is a well-known NP-hard combinatorial optimization problem faced across construction, software, and industrial project management, where the objective is to minimize total project duration while respecting activity precedence relations and limited resource availability. Traditionally, project managers rely on metaheuristic algorithms such as Genetic Algorithms (GA), Simulated Annealing (SA), Bee Colony Optimization (BCO), and priority-rule-based heuristics (PRH) to generate near-optimal schedules; however, no single algorithm consistently outperforms the others across all project types, and running multiple algorithms for every project is computationally expensive and time-consuming. This project proposes a Smart RCPSP Scheduler that automates this decision-making process using a trained neural network classifier. The system accepts project data — activity durations, precedence relationships, and resource requirements — through a structured CSV/manual input interface, and automatically computes three project topology descriptors: Network Complexity (NC), Resource Factor (RF), and Resource Strength (RS). A feedforward neural network, trained on benchmark project instances from the PSPLIB library, uses these descriptors to predict the most suitable scheduling algorithm for the given project. The selected algorithm is then executed to generate an actual project schedule, visualized through an interactive Gantt chart along with resource utilization graphs. Unlike prior work, which stops at algorithm classification, the proposed system extends the approach into a complete, usable scheduling tool with explainable predictions, comparative algorithm evaluation, and generalization across varying project sizes. The system is evaluated on PSPLIB benchmark datasets as well as a real-world construction case study, demonstrating that automated algorithm selection can significantly reduce computational overhead while maintaining schedule quality comparable to exhaustive multi-algorithm evaluation.

Abstract 2

AutoOntology: A Semi-Automatic Framework for Domain Ontology Generation Using Deep Learning and Knowledge Graph Enrichment

The Semantic Web relies on formal ontologies to represent domain knowledge in a machine-readable and reasoning-capable structure, yet many specialized and emerging domains continue to lack such ontologies due to the time-intensive, expertise-dependent nature of manual ontology engineering. This project proposes AutoOntology, a semi-automatic ontology generation system that transforms

unstructured textual documents into a structured, queryable domain ontology with minimal human intervention. The system accepts a corpus of domain-specific documents as input, from which it extracts candidate concepts and keywords using natural language processing techniques, classifies and ranks them by relevance, and enriches them with related entities retrieved from open knowledge graphs such as Wikidata. Semantic similarity measures are then applied to identify and merge overlapping concepts, after which the system constructs formal ontological relationships — including is-a, part-of, super-class-of, and sub-class-of axioms — and validates logical consistency using an OWL reasoner. A distinguishing feature of the proposed system is an interactive human-in-the-loop review interface, through which domain experts can approve, reject, or modify extracted entities and relationships before finalization, making the “semi-automatic” nature of the pipeline both transparent and measurable — an aspect largely unquantified in prior work. The resulting ontology is exported in standard OWL/RDF format, compatible with tools such as Protégé, and is presented to users through an interactive graph-based visualization interface. The framework is evaluated across multiple domains to assess generalizability, and its performance is benchmarked in terms of entity extraction precision, ontology consistency, and the proportion of automatically generated content requiring human correction, offering a more transparent and reproducible alternative to prior closed-API-dependent ontology synthesis approaches.